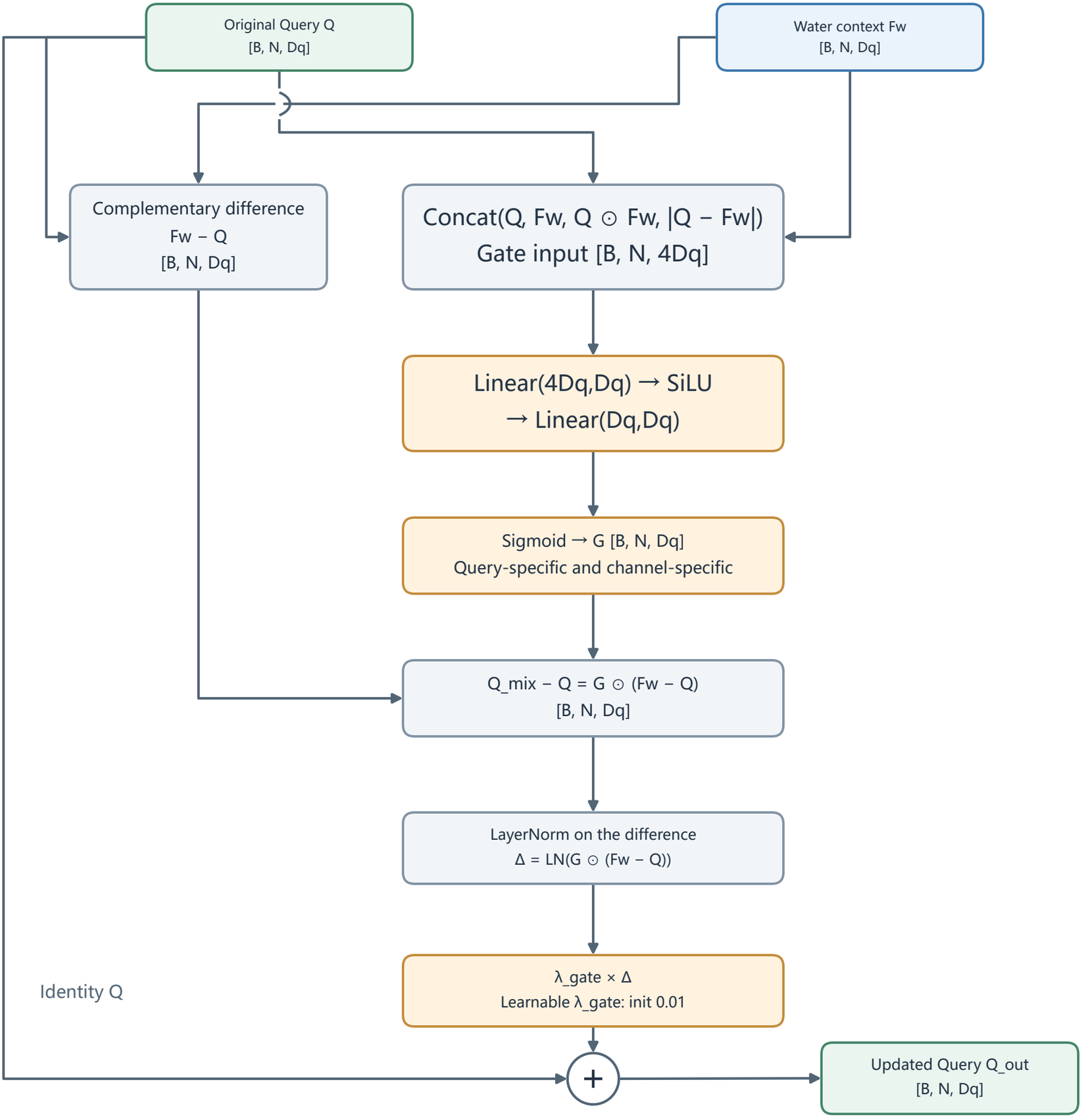
(c) Complementary Query-Water Gate — CQWG

Query 与通道级互补门控 | Residual update of the original object query



Q_mix = (1 − G) ⊙ Q + G ⊙ Fw. The code uses the equivalent difference G ⊙ (Fw − Q).

Q_out = Q + λ_gate × LN(Q_mix − Q). LayerNorm is inside the residual branch, not after the addition.

Each decoder layer learns an independent λ_gate. If λ_gate = 0, Q_out equals Q exactly.